

Real-Time NHL Goal Detection & Smart Light System

Objective

Build a **real-time, locally processed system** that detects when a specific NHL team scores a goal (based on the user's own TV feed) and instantly triggers physical outputs (goal light, LEDs, horn) with minimal delay and no reliance on external APIs.

Core Idea

Instead of relying on delayed NHL data feeds, the system analyzes the **live broadcast video + audio** to detect goals as they appear **on the user's screen**, aiming for near-zero perceived latency.

Hardware Setup

Video Path - HDMI source → HDMI splitter → TV + capture card

- Capture card → USB → laptop

Control Path - Laptop → USB hub → USB relay module

- Relay controls: - Goal light (modified Budweiser light) - LED strips - Optional horn/speakers

Power - Devices powered via wall adapters

- Relay acts as switch (not power source)

Software Architecture (Modular)

1. Video Ingest

- OpenCV reads frames from capture card in real time

2. Audio Ingest

- Monitors audio stream for crowd/horn spikes
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Detection Modules

A. Early Goal Detection (Primary)

- Detect "GOAL" board graphics (earliest signal)
- Not fixed-position (camera moves, occlusion possible)

B. Team Identification (Critical)

- Detect team logo/colors/text within board graphic
- Determines *which team scored* without waiting for scorebug

C. Broadcast Overlay Detection (Fallback)

- Detect TV "GOAL" overlay (more stable but slightly delayed)

D. Scorebug OCR (Support)

- Reads:
 - team labels
 - score
 - period
 - clock
- Used for:
 - mapping team → side
 - replay suppression
 - fallback confirmation

E. Audio Detection (Support)

- Detect crowd/horn/commentator spikes

F. Offensive Zone Estimation (Optional Support)

- Estimate which team was attacking prior to goal
- Used as confidence signal only (not primary decision)

Fusion Logic (Decision Engine)

Trigger ONLY when: - Early goal cue detected (boards or overlay) - Detected team == user-selected team - Audio supports event (optional but helpful) - Event is NOT a duplicate (replay) - System not in cooldown

Replay Protection (Critical)

Store "goal signatures": - period - clock - score

Reject triggers if: - same event appears again (e.g., intermission replay)

Output

When triggered: - Activate relay → goal light + LEDs
- Play goal horn (local audio)
- Optional flashing pattern

Performance Goals

- Detection latency: ~10–50 ms
 - Total system response: ~100–300 ms
 - Must feel **instant relative to the user's broadcast**
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Implementation Stack

- Python
 - OpenCV (video processing)
 - Tesseract OCR
 - Audio processing (basic amplitude detection)
 - USB relay control (serial/HID)
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Key Challenges

- Reliable detection of board graphics (variable position/occlusion)
 - Team identification from graphics/logos
 - OCR stability for scorebug
 - Avoiding false positives (replays, highlights)
 - Tuning fusion thresholds
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Development Strategy

Build in stages:

1. Video feed + relay trigger
2. GOAL overlay detection
3. Scorebug OCR + replay suppression
4. Audio detection
5. Board detection + team identification

6. Fusion logic refinement

Summary

This is a **real-time computer vision + hardware control system** that: - operates fully locally

- syncs perfectly to the user's feed
 - triggers faster than any commercial solution
 - intelligently filters for only the desired team
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Technical Framing

This can be understood as a:

Modular real-time event detection pipeline with hardware actuation
